BIOL 320L 113
GENETICS
Fall 2026

Instructor Information

Patrick Thieringer
Office Location: DN 262
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Office Hours
T 10:00AM-11:30AM | W 2:00-3:30PM

Course Information

BIOL 320L GENETICS
Credit Hours: 4
Meeting Location: DN236
AH312
Meeting Times: R 0930AM-1215PM
MR 200PM-315PM

Course Description

A study of transmission, population, molecular, and cytogenetics. Both in the classroom and the laboratory, the emphasis is on reinforcing basic concepts through a study of the classic experiments in genetics as well as current research. Three-hour lecture, three-hour laboratory per week. Typically offered every semester.

Pre-Requisite/Co-Requisite

PREREQUISITES: CHEM 131-132 or equivalent and a grade of "C" or higher in BIOL 130 & 131, or permission of Instructor

Course Learning Outcomes

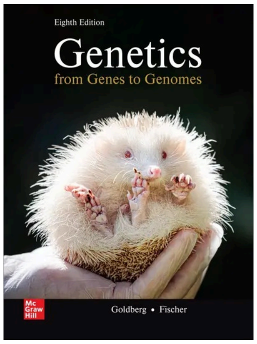
This course introduces the major concepts of Mendelian, molecular, and population genetics. A human perspective will be emphasized particularly concerning the topic of diseases, but we will look at other models as well. One of the goals is that students develop a general understanding of how scientific research works and is carried out. This will be accomplished by students reviewing the history of molecular genetics and the key experiments that were carried out by famous scientists, student presentations for discussion on recent articles published in peer-reviewed journals, and conducting a real, novel research project in the lab whose results could be publishable.

Specific learning objectives will be presented at the start of each lecture. Let these objectives guide your studying and help you to organize your notes, as these are the major topics that will be covered on exams. For exams, you will generally be required to apply this knowledge rather than just memorize lists and facts from lectures and textbook readings.

Overall Course Learning Objectives are listed below:

- CLO1 - Explain the molecular basis of heredity and the flow of genetic information.
- CLO2 - Interpret how gene regulation and epigenetic mechanisms influence an organism's structure, function, and phenotype.
- CLO3 - Analyze patterns of inheritance using Mendelian and more complex genetic models
- CLO4 - Explain how mutation, recombination, chromosome variation, and gene interactions contribute to genetic diversity.
- CLO5 - Interpret and analyze genetic data to draw evidence-based conclusions.
- CLO6 - Evaluate modern applications of genetics and genomics in research, medicine, and biotechnology.
- CLO7 - Explain how foundational discoveries in genetics continue to inform modern biological research and impact society - and how society impacts genetics.

Required Materials



From Genes to Genomes
ISBN: ISBN10: 1265352267 | ISBN13: 9781265352264
Authors: Michael L. Goldberg & Janice Fischer
Publisher: McGraw Hill
Edition: 8th

Optional Resources

We will use Brightspace (https://brightspace.marist.edu/) to organize and distribute content for this class. All announcements will be sent to your Marist email account, so be sure to check this email at least once per day.

Course Grade Calculation

Lecture Components (75%):

Quizzes (5 x 20 pts) - 100 pts

Mid-semester exams (3 x 100 pts) - 300 pts

Final exam - 200 pts

Problem Sets (3 x 25 pts) - 75 pts

Engagement & Participation - 100 pts

Lab Components (25%) :

Lab Practical - 50 pts

Mendelian Modelling Lab Report - 100 pts

CRISPR/Cas Lab Report - 100 pts

Final Course Grade Scale

Grading Scale

Letter Grade	Percentage
A	100 - 95%
A-	94 - 90%
B+	89 - 87%
B	86 - 83%
B-	82 - 80%
C+	79 - 77%
C	76 - 73%
C-	72 - 70%
D+	69 - 65%
D	64 - 60%
F	59% or below

Graded Assignments

LECTURE QUIZZES

There will be five lecture quizzes given throughout the semester. Quizzes will consist of a combination of true/false and multiple choice. Quizzes will be administered in the first 15 minutes of class or on Brightspace. If you have a legitimate excuse for missing a quiz (see Excused Absence Policy), you may make up the missed quiz as long as documentation for the absence is provided. You must contact me via email within 24 hours of the missed quiz to schedule makeups.

LECTURE EXAMS

There will be three midterm exams (non-cumulative), and a final exam for the course that will consist of all of the material from earlier in the semester. All exams will consist of a combination of true/false, multiple-choice, fill- in-the-blank and short answer questions. Exams will cover topics discussed in class and from assigned readings, so your lecture notes and the textbook are recommended study sources. Make-up exams will be given only under exceptional circumstances as described in the Excused Absence Policy section. You must contact me via email within 24 hours of the missed exam to schedule a makeup. Failure to schedule a makeup exam within one week of the exam date will forfeit your points for this exam. Please contact me as soon as possible if you have a conflict with any of the exam dates in this course.

PROBLEM SET ASSIGNMENTS

Problem set assignments in this class are intended to help you study for exams and are due at the beginning of each exam. These assignments MUST BE HANDWRITTEN AND TURNED IN AS WRITTEN COPIES. This can include handwritten homework or written on a tablet device. These assignments can be uploaded to Brightspace, but I will not accept typed out answers to any homework problem. These assignments will include questions like those you may see on exams, including those that help you focus on application of material and concepts learned in each section of the course. You are encouraged to work with peers to complete these assignments and can ask questions about your understanding of concepts needed to complete the assignments during the weekly review sessions, or during office hours, but ALL work must be in your own words.

ENGAGEMENT & PARTICIPATION

Active engagement in class will greatly enhance the course and your success therein. Therefore, this component of your grade (10% of the total), will require not only your consistent attendance, but also participation in discussions, asking and answering questions in class, and completing in-class activities. If you have an unexcused absence, you will forfeit your opportunity for points from that day. Engagement points will not be adversely impacted by excused absences (see Excused Absence Policy)

LATE POLICY

Some assignments (not exams) can be given a grace period from the submission deadline depending on conflicting circumstances. These grace periods need to be communicated and established with me prior to submission.

- Questions about assignments will not be answered during the grace period, or after 5pm on the original due date of the assignment.
- Assignments will NOT be accepted 7 days after the due date and you will receive a zero for that assignment

Following the grace period or time period after the submission date:

- Assignments submitted up to 24 hours after the end of the grace period will incur a 10% penalty.
- An additional 10% penalty will be applied for each additional 24hr period.

Basic Student Responsibilities

You are highly encouraged to attend all class meetings. Quizzes and exams may include material discussed during class that is not contained in the textbook. If you miss a class meeting, you are responsible for learning this material independently; if you miss a class I will be happy to answer specific questions regarding this material during office hours. You are responsible for any announcements made in class regarding changes to the syllabus, or regarding exams, quizzes and/or other assignments.

Excessive unexcused absences will result in a lower engagement grade. If you miss/will miss a class meeting for a legitimate reason (see Excused Absence Policy), please speak with me as soon as possible so that we can try to arrange alternatives.

Communication Policy

You can communicate with me electronically many hours during the week, but we will not spend a lot of time in the same lab or classroom, so I ask that your use of technology in this course remain on-topic. Please do not check your email, shop online, etc. during class time. Inappropriate use of technology during class will result from your ejection from class for that day and loss of engagement and other points.

Please set your phone to silent during class and lab. If you forget and it rings, please silence it as quickly as possible. While laptops/tablets may be used for some in-class/-lab exercises, I prefer that you do not use laptops during lecture components (i.e. to take notes). I may draw on the board, so you should be prepared to draw/diagram content during class (either hard-copy or digitally).

Attendance Policy

EXCUSED ABSENCE POLICY

The following constitute the only valid excuses for missing exams, quizzes, assignments or lab sessions:

1. Death in the immediate family. Present a note from the Center for Advising & Academic Services.

2. Officially-approved College activities such as intercollegiate athletic events or theatrical productions. Present a note from the appropriate College official stating that your participation in this activity prevented you from being in class at the exact date and time it was held.

3. Debilitating illness. Present a note, signed by a physician, stating that the illness prevented you from being in class at the exact date and time that the exam and/or laboratory was held. If you seek medical attention at Marist Health Services (MHS), then you must provide health service with documented authorization allowing me to verify your illness or injury. A note that simply states that you were under the care of a physician, or had an appointment with a doctor, will not be sufficient.

If an absence is excusable and of sudden or unpredictable nature, I still require notification on the day you return to class in order to reschedule missed assignments. Please note that it is your responsibility to schedule a make-up when you return to class. If a make-up is not rescheduled within one week of your return to class, you will have lost your opportunity to do so.

Academic Support

Marist University provides a range of resources to help you get the most from your educational experience. These resources include but are not limited to tutoring, writing assistance, advising, proofreading assistance, and counseling. A full list of these resources can be found at this link. https://www.marist.edu/academic-resources

Technical Support and Support Links

Classrooms and Labs support services are managed through the IT Help Desk and Media Center. Computer related problems should be reported by calling the Marist University IT Help Desk at (845) 575-4357 (xHELP) or email: helpdesk@marist.edu. Problems with overhead projectors or other media related equipment should be reported to the Media Center Help Desk at (845) 575-3635 (x3635) or email: media@marist.edu.

To access Brightspace, visit: https://brightspace.marist.edu/d2l/home and click on "Marist SSO" to login directly with your Marist credentials. For helpful Brightspace tool guides and tutorials, visit: https://my.de.marist.edu/teaching-technologies

Statement on Academic Honesty

Marist University is a learning community dedicated to helping students develop the intellect, characters, and skills required for enlightened, ethical, and productive lives in the global community of the 21st century. Students are expected to pursue excellence in their education while being honest about their work and fair to other members of the learning community. All work presented to instructors for evaluation must reflect their own ideas and effort and must properly acknowledge any contributions of others. Students should expect this honesty and fairness in others as well. As members of the Marist learning community, all students should adhere to the principles of academic integrity as set forth in the Marist Academic Integrity Policy.

https://www.marist.edu/student-life/community/community-standards/academic-integrity-policy

AI Statement

Generative AI tools (e.g., text and image generators like ChatGPT and DALL-E) may not be used to complete assignments unless explicitly permitted. Unauthorized use of AI tools will result in a violation of the academic integrity policy. If AI use is allowed for certain assignments, students must document its role in their work.

Detection Software Statement

Use of software that detects plagiarism, including AI generated content, is mandated in this course. Copyleaks is a service used by Marist University faculty to compare a student's work with its very large database of sources, student papers from other institutions, and the like, to check for originality. Work submitted to Copyleaks will be used only for the purposes of assessing originality, and will not be shared beyond Copyleaks or used for any other purpose. Students must submit all assignments to Copyleaks through regular Brightspace submission process. If your instructor enables Copyleaks, there will be an End User License Agreement (EULA) that you must accept. It will appear to you on the assignment page before the area where you submit the assignment. You can view and accept the EULA by clicking on the link and then clicking "Finish" to accept the terms of use. Students who wish to remove their personal identifying information (name, student identification number, etc.) from the submitted file may do so but must notify their professor ahead of the submission. Work submitted through Brightspace in this course will not be reviewed by the Professor or maintained by the University unless and until the Copyleaks process is completed.

For more information about Copyleaks visit: https://my.de.marist.edu/copyleaks

Accommodations and Accessibility

From the Office of Accommodations and Accessibility

Students with disabilities who believe they may need accommodations in this class are encouraged to contact the Office of Accommodations and Accessibility at (845) 575-3274, Donnelly Hall 226 or via email at accommodations@marist.edu as soon as possible to better ensure that such accommodations are implemented in a timely manner. Marist's guidelines for instructors to comply with the Americans with Disability Act (ADA) are located here: https://www.marist.edu/student-life/community/accommodations-accessibility/guidelines-instructors

Your Privacy Rights - Family Education Rights and Privacy Act

The Family Educational Rights and Privacy Act (FERPA) (20 U.S.C. § 1232g; 34 CFR Part 99) is a Federal law that protects the privacy of student education records. Marist University collects and retains data and information about students for designated periods of time for the expressed purpose of facilitating the student's educational development. The University recognizes the privacy rights of individuals in exerting control over what information about themselves may be disclosed and, at the same time, attempts to balance that right with the institution's need for information relevant to the fulfillment of its educational missions. As employees of the University, faculty, administration and staff all have some access to student record information. Please, visit https://www.marist.edu/academics/registrar/ferpa for more information on how Marist University protects your privacy rights.

Copyright Information

Materials in this course may be subject to copyright protection.

Title IX Statement

Marist University is committed to providing a safe learning environment for all students. If you or someone you know has experienced sexual harassment, including sexual assault, dating or domestic violence, or stalking, support is available. Please contact the Title IX Office at titleix@marist.edu or (845) 575 - 3799 or visit www.marist.edu/title-ix to file a report. Please be aware that faculty and staff are required to disclose incidents of sexual harassment or other potential violations of the Marist University Discrimination, Harassment, and Sexual Misconduct Policy to the Title IX Office. To speak to a confidential resource who does not have this reporting responsibility, contact Counseling Services at (845) 575 - 3314, Health Services at (845) 575 - 3270, or Department of Spiritual Life & Service at (845) 575 - 3000 (x2275). For more information about reporting options and resources at Marist University, please visit https://www.marist.edu/title-ix

Community and Belonging Statement

The university's academic mission is immeasurably enriched by students with diverse experiences. Our finest efforts as intellectual beings heavily rely on the exchange of ideas. Interactions in our classrooms among persons and groups with diverse backgrounds, ideologies, and experiences facilitate these efforts by allowing us all to be more reflective about the varied historical and social contexts in which we work and learn. For faculty and students to continue being leaders inside and beyond academia, we must ensure that we consider the diversity of all who comprise our communities and foster a climate in which those diverse influences are respected and valued. In this course, we will challenge each other's thinking while working collaboratively to ensure that the classroom is a space of safety and bravery. Our classroom offers an environment where individuals of varying opinions, experiences, and backgrounds are able to be free to learn without fear of being silenced. Evidence of these efforts will manifest in readings, lectures/class discussion, seminars, and group projects. Aspects of diversity include, but are not limited to, race, ethnicity, color, nationality, sex, gender, gender identity, gender expression, class, sexual orientation, religion, age, ability, and veteran status. Students who would like to be identified in a manner other than what is indicated on the course roster can contact me privately via phone, email, web conference or face-to-face meeting to indicate name, pronoun and any other preferences they may have.

Laboratory - Specific Information, Descriptions, and Policies

Goals and Learning Objectives

The laboratory component of Genetics is designed to provide you with opportunities to allow you to practice molecular and mendelian genetic techniques. Laboratory exercises are designed to both reinforce and expand upon topics discussed in lecture; moreover, the labs offer hands-on experience and provide an opportunity to develop laboratory skills that will be essential to future coursework as well as provide experience with materials and techniques that carry over to post-graduate work.

Evaluations

The goals of the laboratory component of the course will be assessed through performance on practical technique, written assignments, and lab reports, as well as active participation. Details on these assignments, including instructions and whether they should be submitted electronically or during lab, will be provided during the laboratory sessions and will also be posted on Brightspace. Participation points will be given for each lab based on lab attendance, active participation in the lab activities with lab partners, and engagement with the material through taking notes, asking questions, etc. There will be no written exams for the laboratory component of the course.

Laboratory Points Breakdown (25% of total grade):

Lab Practical - 50 pts

Mendelian Modeling Lab Report - 100 pts

CRISPR Lab Report - 100 pts

Total Points - 250 pts

Laboratory Safety

The primary concern of the Marist faculty and administration regarding laboratory exercises is that they be conducted safely. Laboratory experiments may involve the use of hazardous materials including, but not limited to, sharp objects, concentrated acids and bases, and mutagenic and carcinogenic compounds. You should always act in a manner that will minimize the chance of injury should an accident occur. Likewise, you should always assume that an accident can occur and follow appropriate guidelines designed to protect you in the event of an accident. Always follow the safety advice given to you by your TA or instructor. The following are rules that you should follow to protect yourself:

1. Drinking and eating is not allowed.

2. Open-toed shoes are not allowed.

3. Short pants and cropped/sleeveless shirts are not allowed.

4. Use protective eyewear whenever requested to do so.

Laboratory Schedule

Please note that the laboratory schedule is subject to change at my discretion – any changes made to the schedule will be announced in class and/or on Brightspace.

Course Calendar

COURSE CALENDAR - Subject to Change

Week	Date	Topic	Required Reading	Assessments	Homework
1	31-Aug	DNA Discovery & Structure	Chapter 6 (6.1 - 6.2)		
	3-Sep	DNA Replication & Telomeres	Chapter 6 (6.3 - 6.4), Chapter 13 (13.4)		Week 1 HW Problems
2	7-Sep	Labor Day - No Class			
	10-Sep	Gene Expression: Transcription	Chapter 9 (9.1)	**Quiz 1**	Week 2 HW Problems
3	14-Sep	Gene Expression: Translation	Chapter 9 (9.2)		
	17-Sep	Effects of Mutations on Gene Expression	Chapter 9 (9.4)		Week 3 HW Problems
4	21-Sep	DNA Analysis & Sequencing	Chapter 10 (10.1-10.4), Chapter 12 (12.1-12.5)	**Quiz 2**	
	24-Sep	Epigenetics (Molecular Gene Expression)	Chapter 20 (20.1)		Week 4 HW Problems
5	28-Sep	Exam 1 Review			
	1-Oct	**Exam 1**		Week 1-4 HW Problem Sets	
6	5-Oct	Chromosomal Basis of Inheritance	Chapter 3, Chapter 4 (4.2), Chapter 15 (15.1)		
	8-Oct	Mendelian Genetics I (Single Gene)	Chapter 1		Week 6 HW Problems
7	12-Oct	Mendelian Genetics II (Dihybrid Crosses)	Chapter 1		
	15-Oct	Mendelian Genetics III (Inheritance of Complex Traits)	Chapter 1	**Quiz 3**	Week 7 HW Problems
8	19-Oct	Extensions to Mendel's Laws	Chapter 2		
	22-Oct	Genetic Linkage & Mapping	Chapter 5 (5.1-5.4)		Week 8 HW Problems
9	26-Oct	Genetics of Complex Traits	Chapter 25		
	29-Oct	Epigenetics (Imprinting)	Chapter 20 (20.2)	**Quiz 4**	Week 9 HW Problems
10	2-Nov	Exam 2 Review			
	5-Nov	**Exam 2**		Week 6-9 HW Problem Sets	
11	9-Nov	Mutations & DNA Repair Mechanisms	Chapter 7		
	12-Nov	Cancer Genetics I	Chapter 23		Week 11 HW Problems
12	16-Nov	Cancer Genetics II	Chapter 23		
	19-Nov	Genetic Engineering	Chapter 21		Week 12 HW Problems
13	23-Nov	Population Genetics I	Chapter 24		
	26-Nov	Population Genetics II	Chapter 24	**Quiz 5**	Week 13 HW Problems
14	30-Nov	Exam 3 Review			
	3-Dec	**Exam 3**		Week 11-13 HW Problem Sets	
15	7-Dec	Special Topic in Genetics			
	10-Dec	Final Exam Review			
	14-Dec	**Final Exam @ 1:00 PM**			

LABORATORY CALENDAR - Subject to Change

Week	Date	Topic	Assignment	Required Readings
1	3-Sep	Introductions & Orientations		Lab Protocol
2	10-Sep	Sequencing - Part 1	Lab Safety Acknowledgement Signature	Lab Protocol
3	17-Sep	Sequencing - Part 2		Lab Protocol
4	24-Sep	Sequencing - Part 3		Lab Protocol
5	1-Oct	Sequencing - Part 4		Lab Protocol
6	8-Oct	Lab Practical		Lab Protocol
7	15-Oct	Mendelian Inheritance - Part 1		Lab Protocol
8	22-Oct	Mendelian Inheritance - Part 2		Lab Protocol
9	29-Oct	Mendelian Inheritance - Part 3		Lab Protocol
10	5-Nov	Mendelian Inheritance - Part 4		Lab Protocol
11	12-Nov	CRISPR Lab - Part 1	Mendelian Inheritance Lab Report	Lab Protocol
12	19-Nov	CRISPR Lab - Part 2		Lab Protocol
13	26-Nov	CRISPR Lab - Part 3		Lab Protocol
14	3-Dec	CRISPR Lab - Part 4		Lab Protocol
15	10-Dec	Final Exam Review	CRISPR Lab Report	Lab Protocol